

Target Project

source tree | build/test commands | compile DB

Security-Aware Context Graph

functions/blocks + call/control/data/resource edges
error paths, return summaries, side effects

Opportunity Detection

goto/resource/duplicate/dead/smell detectors
severity, confidence, evidence, ranking

Safety Contract

affected files/functions, invariants,
security constraints, validation strategy

LLM Patch Candidates

JSON edit plan or unified diff
conservative/aggressive generation modes

Validation Gates

apply -> build/tests -> static analysis
resource -> semantic -> security checks

Report

selected patch | failed candidates | traces

feedback